Dear Authors: Review of the Manuscript TNNLS-2026-P-48775

Paper title: Constructive and Destructive Training Regimes: A Framework for Interpreting SGD Stability

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Dear Authors,

Summary. This manuscript aims to study how temporal cross-step interference affects SGD training dynamics by reinforcing updates in a constructive regime or partially canceling updates in a destructive regime. The manuscript argues that the gradient alignment of two consecutive mini-batches, defined in Eq. (13) as

$$\rho_t = \frac{G_{B_{t-1}}^\top G_{B_t}}{\|G_{B_{t-1}}\|_2 \|G_{B_t}\|_2},$$

is a practical diagnostic indicating whether successive updates contribute to constructive or destructive regimes. This Equ (13) is obtained by expanding

$$\|\theta_T\|_2^2 = \left\| \theta_0 - \eta \sum_{t=0}^{T} G_{B_t} \right\|_2^2,$$

while assuming

$$\left\| G_s^\top G_t \right\| \to 0, \qquad |t-s| > 1.$$

This analysis framework (by expanding the ℓ_2-norm of parameter vector over the training trajectory from time point 0 to time point T is applied to analyze the stability problem — when the update cancels the signal or reinforce the signal.

Major concerns

C1. Stability framing. The manuscript should clarify its notion of "stability." Although it reviews classical curvature-based stability, the proposed notion is informal and seems closer to update-level interference or catastrophic-forgetting-style cancellation. I recommend that the authors define this notion precisely and distinguish it from standard curvature-based stability; below, I use quoted "stability" to refer to the manuscript's intended notion.

C2. Connection parameter-vector's 2-norm to "stability". There lacks a major justification/connection in that how the 2-norm $\|\theta_T\|$ links to the "stability". Without a rigorously theoretical or empirical justification, the central arguments in this manuscript no longer

holds true.

The manuscript starts from the 2-norm of the final parameter vector θ_T

$$\|\theta_T\|_2^2 = \left\| \theta_0 - \eta \sum_{t=0}^{T} G_{B_t} \right\|_2^2,$$

to argue that the cross term

$$G_{B_{t-1}}^\top G_{B_t}, \tag{1}$$

is crucial to the improvement of the 2-norm of parameter vector by assuming

$$\left\| G_s^\top G_t \right\| \to 0, \qquad |t-s| > 1.$$

C3. Strong assumption without empirical justification. To defend the argument that the alignment of the gradients of consecutive batches matter, the manuscript imposes a strong assumption:

$$\left\| G_s^\top G_t \right\| \to 0, \qquad |t-s| > 1.$$

However there is no evidence to support this claim. Without experimental or theoretical justification, the claims do not hold in this manuscript.

C4. Statement in Section II-B on mini-batch stochasticity. The manuscript states

> *"This makes clear why $\Sigma(\theta_t)$ is generally not a viable online diagnostic: computing it requires access to the full gradient $g^{(t)}$ and all per-sample gradients $\{g_i^{(t)}\}_{i=1}^N$ at each training step, which is impractical and incurs at least a full-pass cost per step."*

This statement is too strong. For negative log-likelihood losses, including cross-entropy, the gradient covariance is closely related to the Fisher information. Let $\ell(\theta; z) = -\log p_\theta(z)$ and $s_\theta(z) = \nabla_\theta \log p_\theta(z)$. Then $\nabla_\theta \ell(\theta; z) = -s_\theta(z)$, and under the usual regularity condition,

$$\mathbb{E}_{z\sim p_\theta}[s_\theta(z)] = \nabla_\theta \int p_\theta(z)\, \mathrm{d}z = \nabla_\theta 1 = 0.$$

Hence,

$$\mathrm{Cov}(\nabla_\theta \ell(\theta; z)) = \mathbb{E}[s_\theta(z) s_\theta(z)^\top] = F(\theta),$$

where $F(\theta)$ is the Fisher information matrix. Thus, while the exact full-dataset empirical covariance is expensive to compute, Fisher-type quantities admit stochastic mini-batch estimates and online approximations.

Sincerely,
Reviewer